

Лабораторная работа №3

Основы информационной безопасности

Лисенков Е.Р.

Российский университет дружбы народов, Москва, Россия

Информация

- Лисенков Егор Романович
- студент
- Российский университет дружбы народов
- 1132232881@rudn.ru
- <https://github.com/erlisenkov>



Вводная часть

Получение практических навыков работы в консоли с атрибутами файлов для групп пользователей¹.

Задания

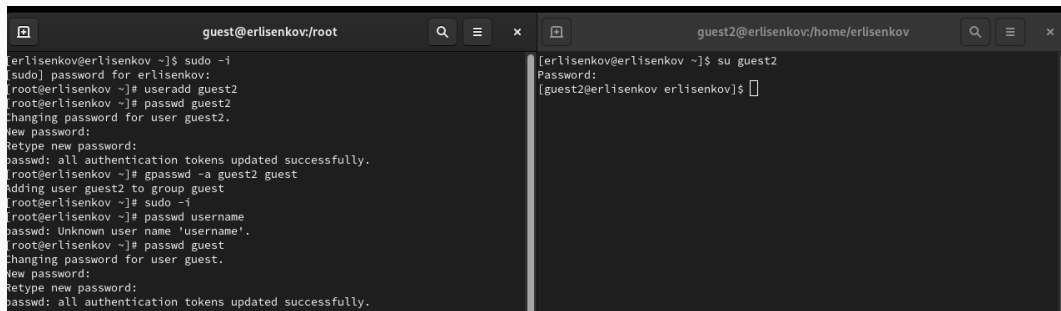
Научиться работать с консолью Rocky Linux.

Создаю дополнительного
пользователя (рис.1).

Создаю дополнительного пользователя (рис.1).

```
[erlisenkov@erlisenkov ~]$ sudo -i
[sudo] password for erlisenkov:
[root@erlisenkov ~]# useradd guest2
[root@erlisenkov ~]# passwd guest2
Changing password for user guest2.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@erlisenkov ~]# gpasswd -a guest2 guest
Adding user guest2 to group guest
[root@erlisenkov ~]#
```

Подтверждение (рис.2)

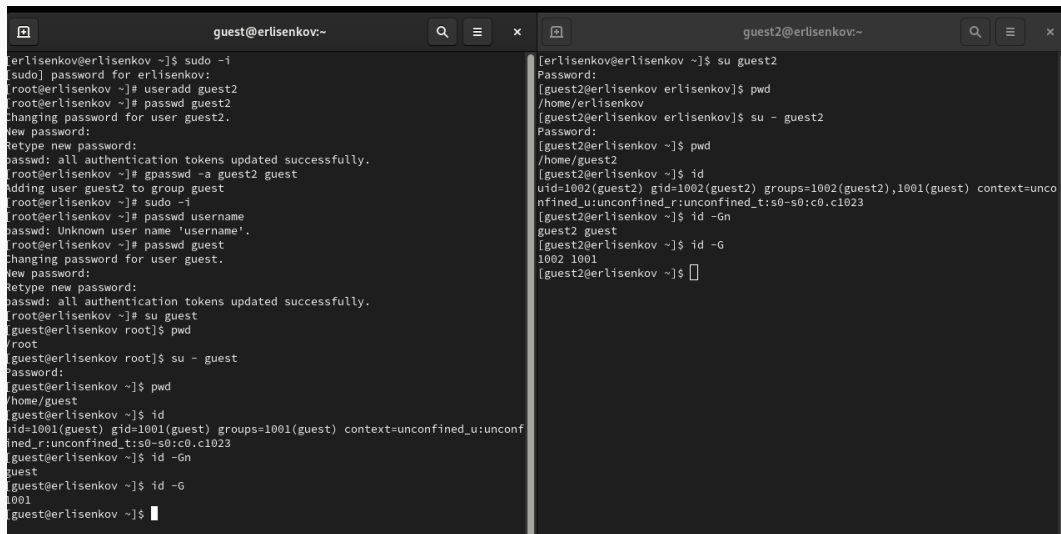


The image shows two terminal windows side-by-side. The left window is titled 'guest@erlisenkov:/root' and shows a sequence of commands to create a user and set passwords. The right window is titled 'guest2@erlisenkov:/home/erlisenkov' and shows the user switching to 'guest2' and logging in.

```
guest@erlisenkov:/root
[erlisenkov@erlisenkov ~]$ sudo -i
[sudo] password for erlisenkov:
[root@erlisenkov ~]# useradd guest2
[root@erlisenkov ~]# passwd guest2
Changing password for user guest2.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@erlisenkov ~]# gpasswd -a guest2 guest
Adding user guest2 to group guest
[root@erlisenkov ~]# sudo -i
[root@erlisenkov ~]# passwd username
passwd: Unknown user name 'username'.
[root@erlisenkov ~]# passwd guest
Changing password for user guest.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.

guest2@erlisenkov:/home/erlisenkov
[erlisenkov@erlisenkov ~]$ su guest2
Password:
[guest2@erlisenkov erlisenkov]$
```

Начинаю проверку системы guest2 (рис. 3)



```
guest@erlisenkov:~  
[erlisenkov@erlisenkov ~]$ sudo -i  
[sudo] password for erlisenkov:  
[root@erlisenkov ~]# useradd guest2  
[root@erlisenkov ~]# passwd guest2  
Changing password for user guest2.  
New password:  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@erlisenkov ~]# gpasswd -a guest2 guest  
Adding user guest2 to group guest  
[root@erlisenkov ~]# sudo -i  
[root@erlisenkov ~]# passwd username  
passwd: Unknown user name 'username'.  
[root@erlisenkov ~]# passwd guest  
Changing password for user guest.  
New password:  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@erlisenkov ~]# su guest  
[guest@erlisenkov root]$ pwd  
/root  
[guest@erlisenkov root]$ su - guest  
Password:  
[guest@erlisenkov ~]$ pwd  
/home/guest  
[guest@erlisenkov ~]$ id  
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[guest@erlisenkov ~]$ id -Gn  
guest  
[guest@erlisenkov ~]$ id -G  
1001  
[guest@erlisenkov ~]$ █  
  
guest2@erlisenkov:~  
[erlisenkov@erlisenkov ~]$ su guest2  
Password:  
[guest2@erlisenkov erlisenkov]$ pwd  
/home/erlisenkov  
[guest2@erlisenkov erlisenkov]$ su - guest2  
Password:  
[guest2@erlisenkov ~]$ pwd  
/home/guest2  
[guest2@erlisenkov ~]$ id  
uid=1002(guest2) gid=1002(guest2) groups=1002(guest2),1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[guest2@erlisenkov ~]$ id -Gn  
guest2 guest  
[guest2@erlisenkov ~]$ id -G  
1002 1001  
[guest2@erlisenkov ~]$ █
```

Проверка ID (рис.4)

```
changing password for user guest.  
New password:  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@erlisenkov ~]# su guest  
[guest@erlisenkov root]$ pwd  
/root  
[guest@erlisenkov root]$ su - guest  
Password:  
[guest@erlisenkov ~]$ pwd  
/home/guest  
[guest@erlisenkov ~]$ id  
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
[guest@erlisenkov ~]$ id -Gn  
guest  
[guest@erlisenkov ~]$ id -G  
1001  
[guest@erlisenkov ~]$ groups guest  
guest : guest  
[guest@erlisenkov ~]$
```

```
1002 1001  
[guest2@erlisenkov ~]$ groups guest2  
guest2 : guest2 guest  
[guest2@erlisenkov ~]$
```

Сравнение (рис.5).

```
/root
[guest@erlisenkov root]$ su - guest
Password:
[guest@erlisenkov ~]$ pwd
/home/guest
[guest@erlisenkov ~]$ id
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[guest@erlisenkov ~]$ id -Gn
guest
[guest@erlisenkov ~]$ id -G
1001
[guest@erlisenkov ~]$ groups guest
guest : guest
[guest@erlisenkov ~]$ cat /etc/group | grep "guest"
guest:x:1001:guest2
guest2:x:1002:
[guest@erlisenkov ~]$ chmod g+rwX /home/guest
[guest@erlisenkov ~]$ chmod 000 dir1
[guest@erlisenkov ~]$ ls -l
total 0
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Desktop
d----- . 2 guest guest 19 Feb 23 18:12 dir1
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Documents
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Downloads
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Music
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Pictures
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Public
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Templates
drwxr-xr-x. 2 guest guest 6 Feb 23 17:53 Videos
[guest@erlisenkov ~]$
```

```
[guest2@erlisenkov ~]$ id -Gn
guest2 guest
[guest2@erlisenkov ~]$ id -G
1002 1001
[guest2@erlisenkov ~]$ groups guest2
guest2 : guest2 guest
[guest2@erlisenkov ~]$ cat /etc/group | grep "guest2"
guest:x:1001:guest2
guest2:x:1002:
[guest2@erlisenkov ~]$ newgrp guest
[guest2@erlisenkov ~]$
```

Тестирую функции прав (рис.6)

```
d---rwx---x. 2 guest guest 19 Sep 13 19:25 dir1
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Documents
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Downloads
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Music
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Pictures
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Public
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Templates
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Videos
[guest@dbelicheva ~]$ chmod 770 dir1
[guest@dbelicheva ~]$ ls -l
ls: cannot access 'l': No such file or directory
[guest@dbelicheva ~]$ ls -l
total 0
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Desktop
drwxrwx---. 2 guest guest 19 Sep 13 19:25 dir1
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Documents
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Downloads
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Music
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Pictures
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Public
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Templates
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Videos
[guest@dbelicheva ~]$ cd dir1
[guest@dbelicheva dir1]$ ls -l
total 0
-----x. 1 guest guest 0 Sep 13 19:25 file2
[guest@dbelicheva dir1]$ chmod 010 file2
[guest@dbelicheva dir1]$ ls -l
total 0
-----x---. 1 guest guest 0 Sep 13 19:25 file2
[guest@dbelicheva dir1]$ chmod 070 file2
[guest@dbelicheva dir1]$ ls -l
total 0
---rwx---. 1 guest guest 0 Sep 13 19:25 file2
[guest@dbelicheva dir1]$ ls -l
total 4
-rw-r--r--. 1 guest2 guest 5 Sep 20 20:36 file1
---rwx---. 1 guest guest 0 Sep 13 19:25 file2
[guest@dbelicheva dir1]$ cd ..
[guest@dbelicheva ~]$ ls -l
total 0
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Desktop
drwxrwx---. 2 guest guest 32 Sep 20 20:35 dir1
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Documents
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Downloads
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Music
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Pictures
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Public
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Templates
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Videos
[guest@dbelicheva ~]$
```

```
[guest2@dbelicheva guest]$ cd dir1
[guest2@dbelicheva dir1]$ ls -l
total 0
-----x. 1 guest guest 0 Sep 13 19:25 file2
[guest2@dbelicheva dir1]$ ls
file2
[guest2@dbelicheva dir1]$ cd ..
[guest2@dbelicheva guest]$ ls -l
total 0
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Desktop
drwxrwx---. 2 guest guest 19 Sep 13 19:25 dir1
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Documents
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Downloads
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Music
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Pictures
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Public
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Templates
drwxr-xr-x. 2 guest guest 6 Sep 13 18:45 Videos
[guest2@dbelicheva guest]$ cd dir1
[guest2@dbelicheva dir1]$ ls -l
total 0
-----x---. 1 guest guest 0 Sep 13 19:25 file2
[guest2@dbelicheva dir1]$ chmod 000 file2
chmod: changing permissions of 'file2': Operation not p
ermitted
[guest2@dbelicheva dir1]$ touch file1
[guest2@dbelicheva dir1]$ ls -l
total 0
-rw-r--r--. 1 guest2 guest 0 Sep 20 20:35 file1
---rwx---. 1 guest guest 0 Sep 13 19:25 file2
[guest2@dbelicheva dir1]$ echo "test" > file1
[guest2@dbelicheva dir1]$ cat file1
test
[guest2@dbelicheva dir1]$ touch file3
[guest2@dbelicheva dir1]$ ls -l
total 4
-rw-r--r--. 1 guest2 guest 5 Sep 20 20:36 file1
---rwx---. 1 guest2 guest 5 Sep 20 20:36 file1
---rwx---. 1 guest guest 0 Sep 13 19:25 file2
-rw-r--r--. 1 guest2 guest 0 Sep 20 20:39 file3
[guest2@dbelicheva dir1]$ echo "test" > file3
[guest2@dbelicheva dir1]$ cat file3
test
[guest2@dbelicheva dir1]$ chmod 000 file1
[guest2@dbelicheva dir1]$ chmod 000 file2
chmod: changing permissions of 'file2': Operation not p
ermitted
[guest2@dbelicheva dir1]$ chmod 100 file2
chmod: changing permissions of 'file2': Operation not p
ermitted
[guest2@dbelicheva dir1]$
```

Выводы

Я усвоил материал и готов к дальнейшему изучению линукс!